JAVA ASSIGNMENT 04

Q1.

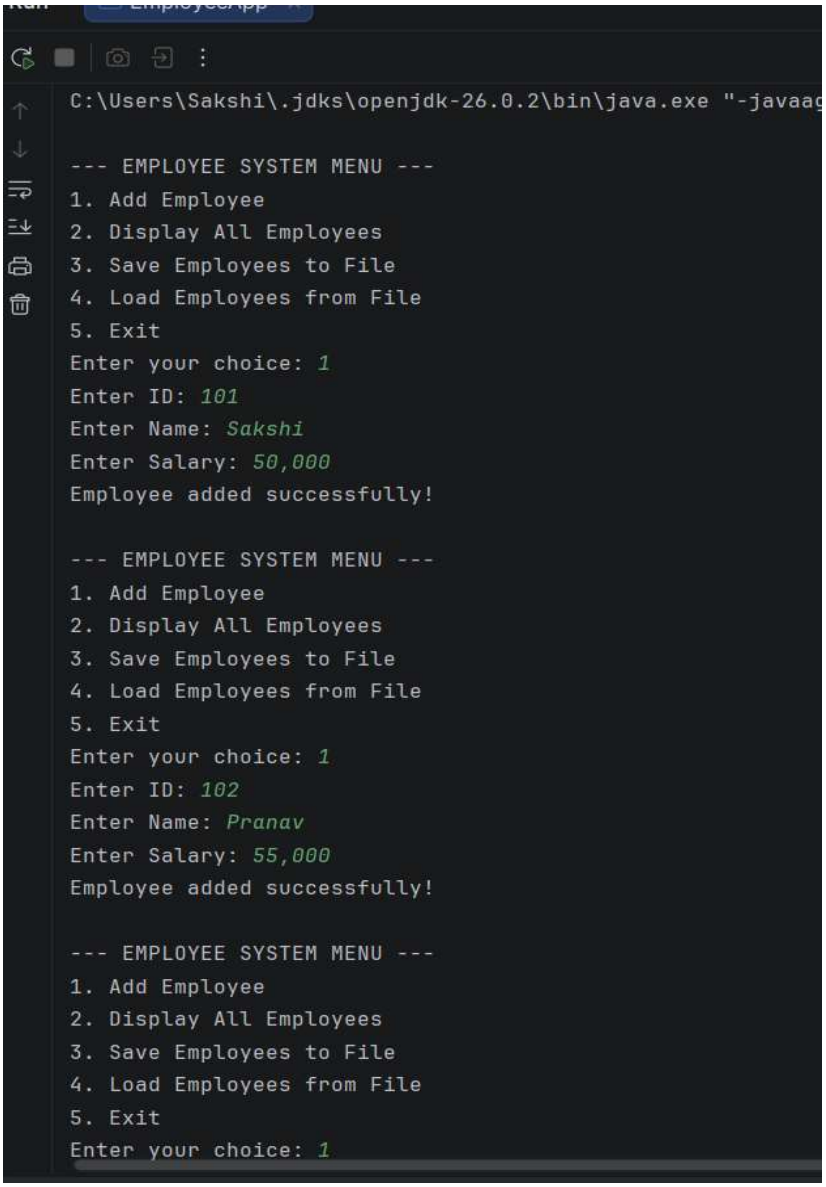
CODE:

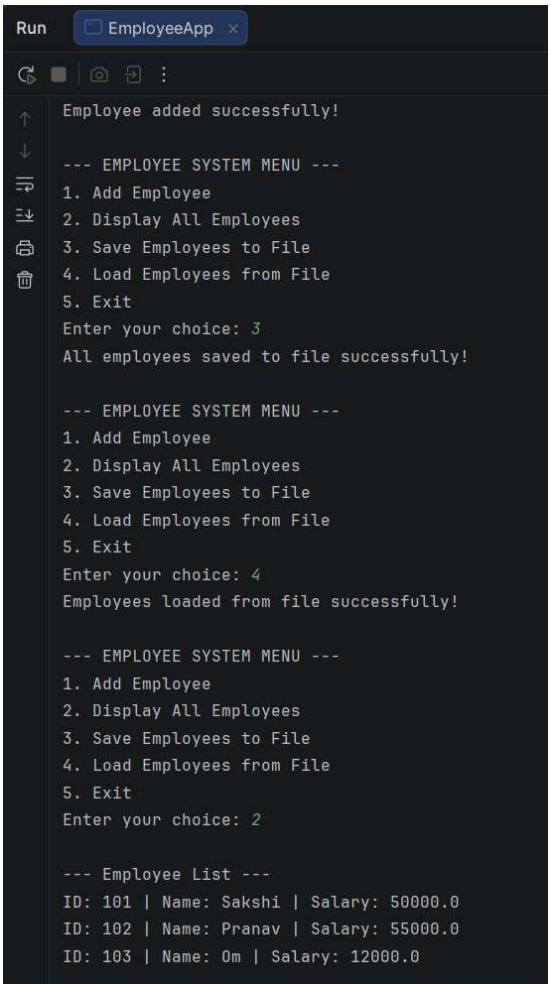
```
import java.io.*;
import java.util.Scanner;
class Employee implements Serializable {
   private static final long serialVersionUID = 1L;
   int id;
   String name;
   double salary;
   public Employee(int id, String name, double salary) {
      this.id = id;
      this.name = name;
      this.salary = salary;
   }
   public void display() {
      System.out.println("ID: " + id + " | Name: " + name + " | Salary: " +
salary);
   }
}
public class EmployeeApp {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      Employee[] employees = new Employee[10];
      int count = 0;
      String fileName = "employees.dat";
      while (true) {
         System.out.println("\n--- EMPLOYEE SYSTEM MENU ---");
         System.out.println("1. Add Employee");
         System.out.println("2. Display All Employees");
         System.out.println("3. Save Employees to File");
         System.out.println("4. Load Employees from File");
         System.out.println("5. Exit");
         System.out.print("Enter your choice: ");
```

```
int choice = sc.nextInt();
sc.nextLine();
if (choice == 1) {
   if (count < employees.length) {
      System.out.print("Enter ID: ");
      int id = sc.nextInt();
      sc.nextLine();
      System.out.print("Enter Name: ");
      String name = sc.nextLine();
      System.out.print("Enter Salary: ");
      double salary = sc.nextDouble();
      employees[count] = new Employee(id, name, salary);
      count++;
      System.out.println("Employee added successfully!");
   } else {
      System.out.println("Array is full! Cannot add more employees.");
   }
} else if (choice == 2) {
   if (count == 0) {
      System.out.println("No employees found in memory.");
   } else {
      System.out.println("\n--- Employee List ---");
      for (int i = 0; i < count; i++) {
         employees[i].display();
      }
   }
} else if (choice == 3) {
   try {
      FileOutputStream fos = new FileOutputStream(fileName);
      ObjectOutputStream oos = new ObjectOutputStream(fos);
      oos.writeInt(count);
      for (int i = 0; i < count; i++) {
         oos.writeObject(employees[i]);
      }
      oos.close();
      fos.close();
      System.out.println("All employees saved to file successfully!");
```

```
            } catch (IOException e) {
                System.out.println("Error saving to file: " + e.getMessage());
            }
        } else if (choice == 4) {
            try {
                FileInputStream fis = new FileInputStream(fileName);
                ObjectInputStream ois = new ObjectInputStream(fis);
                count = ois.readInt();
                for (int i = 0; i < count; i++) {
                    employees[i] = (Employee) ois.readObject();
                }
                ois.close();
                fis.close();
                System.out.println("Employees loaded from file successfully!");
            } catch (FileNotFoundException e) {
                System.out.println("No saved file found. Please save data first.");
            } catch (IOException | ClassNotFoundException e) {
                System.out.println("Error loading from file: " + e.getMessage());
            }
        } else if (choice == 5) {
            System.out.println("Exiting program. Goodbye!");
            break;
        } else {
            System.out.println("Invalid choice! Please try again.");
        }
    }
    sc.close();
  }
}
```

OUTPUT:


C:\Users\Sakshi\.jdks\openjdk-26.0.2\bin\java.exe "-javaag
--- EMPLOYEE SYSTEM MENU ---
1. Add Employee
2. Display All Employees
3. Save Employees to File
4. Load Employees from File
5. Exit
Enter your choice: 1
Enter ID: 101
Enter Name: Sakshi
Enter Salary: 50,000
Employee added successfully!
--- EMPLOYEE SYSTEM MENU ---
1. Add Employee
2. Display All Employees
3. Save Employees to File
4. Load Employees from File
5. Exit
Enter your choice: 1
Enter ID: 102
Enter Name: Pranav
Enter Salary: 55,000
Employee added successfully!
--- EMPLOYEE SYSTEM MENU ---
1. Add Employee
2. Display All Employees
3. Save Employees to File
4. Load Employees from File
5. Exit
Enter your choice: 1




Q2.

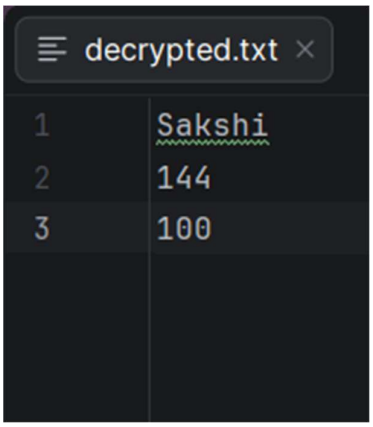
CODE:

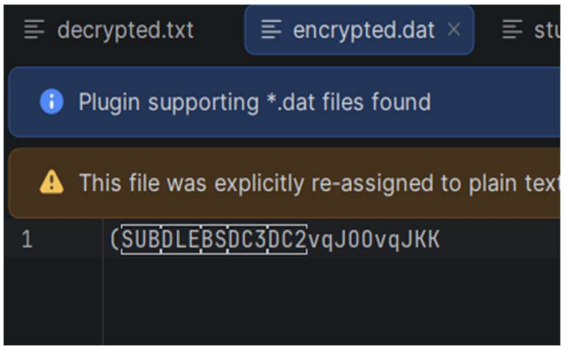
```
package Assingemnt3;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;
import java.util.Scanner;
public class Program{
   static final int KEY=123;
   static final String FOLDER= "C:\\Users\\Sakshi\\IdeaProjects\\CDAC
Assignment 2\\Assignment3\\src\\";
   public static void processFile(String inputFile, String outputFile) {
      System.out.println("Input File: "+inputFile);
```

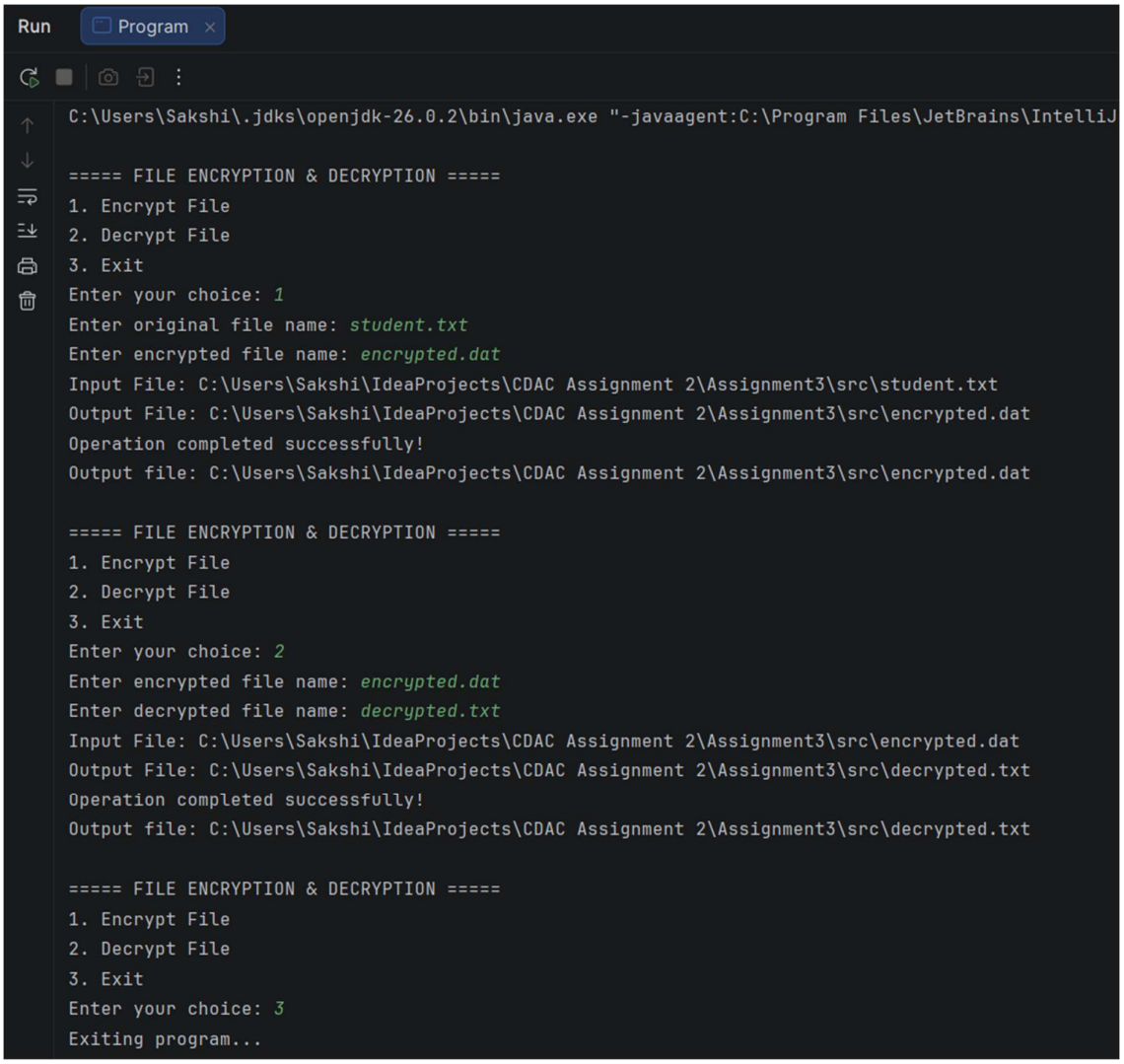
```
        System.out.println("Output File: "+outputFile);
        try (
                FileInputStream fis = new FileInputStream(inputFile);
                FileOutputStream fos = new FileOutputStream(outputFile)
        ) {
            int data;
            while ((data = fis.read()) != -1) {
                int result = data ^ KEY;
                fos.write(result);
            }
            System.out.println("Operation completed successfully!");
            System.out.println("Output file: " + outputFile);
        } catch (IOException e){
            System.out.println("Error: " + e.getMessage());
        }
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("\n===== FILE ENCRYPTION & DECRYPTION
=====");
            System.out.println("1. Encrypt File");
            System.out.println("2. Decrypt File");
            System.out.println("3. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            sc.nextLine();
            switch (choice) {
                case 1:
                    System.out.print("Enter original file name: ");
                    String originalFile = sc.nextLine().trim();
                    System.out.print("Enter encrypted file name: ");
                    String encryptedFile = sc.nextLine().trim();
                    processFile(
                            FOLDER + originalFile,
                            FOLDER + encryptedFile
```

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                );
                break;
            case 2:
                System.out.print("Enter encrypted file name: ");
                String encryptedInput = sc.nextLine();
                System.out.print("Enter decrypted file name: ");
                String decryptedFile = sc.nextLine();
                processFile(
                        FOLDER + encryptedInput,
                        FOLDER + decryptedFile
                );
                break;
            case 3:
                System.out.println("Exiting program...");
                break;
            default:
                System.out.println("Invalid choice!");
        }
    } while (choice != 3);
    sc.close();
  }
}
```

OUTPUT:




Run
Program
C:\Users\Sakshi\.jdks\openjdk-26.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ
===== FILE ENCRYPTION & DECRYPTION =====
1. Encrypt File
2. Decrypt File
3. Exit
Enter your choice: 1
Enter original file name: student.txt
Enter encrypted file name: encrypted.dat
Input File: C:\Users\Sakshi\IdeaProjects\CDAC Assignment 2\Assignment3\src\student.txt
Output File: C:\Users\Sakshi\IdeaProjects\CDAC Assignment 2\Assignment3\src\encrypted.dat
Operation completed successfully!
Output file: C:\Users\Sakshi\IdeaProjects\CDAC Assignment 2\Assignment3\src\encrypted.dat
===== FILE ENCRYPTION & DECRYPTION =====
1. Encrypt File
2. Decrypt File
3. Exit
Enter your choice: 2
Enter encrypted file name: encrypted.dat
Enter decrypted file name: decrypted.txt
Input File: C:\Users\Sakshi\IdeaProjects\CDAC Assignment 2\Assignment3\src\encrypted.dat
Output File: C:\Users\Sakshi\IdeaProjects\CDAC Assignment 2\Assignment3\src\decrypted.txt
Operation completed successfully!
Output file: C:\Users\Sakshi\IdeaProjects\CDAC Assignment 2\Assignment3\src\decrypted.txt
===== FILE ENCRYPTION & DECRYPTION =====
1. Encrypt File
2. Decrypt File
3. Exit
Enter your choice: 3
Exiting program...